

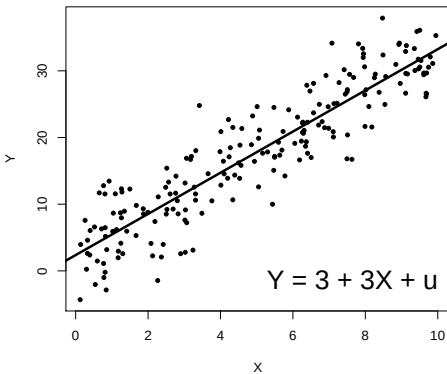
PLSC 476: Empirical Legal Studies

Christopher Zorn

October 9, 2025

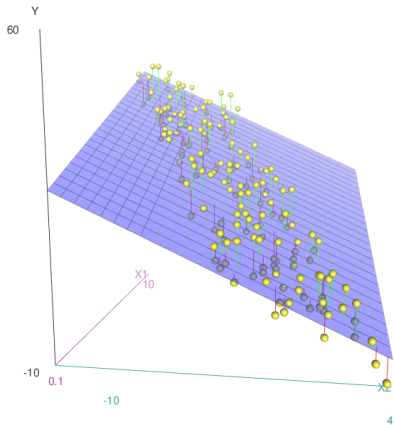
(Bivariate) Regression Redux

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$



Multiple Regression

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$



Multiple Regression

Formally:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots \beta_k X_{ki} + u_i$$

Interpretation

- $\beta_k > 0$: *positive marginal association* between X_k and Y
- $\beta_k < 0$: *negative marginal association* between X_k and Y
- *Value of β_k* : “All else equal, a one-unit change in X_k is associated with an expected β_k -unit difference in Y .”
- When X_k is dichotomous (0 or 1): “All else equal, the presence of X_k changes the expected value of Y by β_k units.”

Assumptions:

- *Linearity*
- *Additivity*
- Other technical assumptions...

When $Y \in \{0, 1\}$ (*binary*):

- $E(Y) = \text{Prob}(Y = 1)$
- *Linear Probability Model*
- Changes in X are associated with changes in the *probability* that $Y = 1$

The Issue: *Habeas Corpus*

- Common law: “The Great Writ”
- What is it?
 - “A writ of habeas corpus is used to bring a prisoner or other detainee (e.g., institutionalized mental patient) before the court to determine if the person’s imprisonment or detention is lawful. A habeas petition proceeds as a civil action against the State agent (usually a warden) who holds the defendant in custody.” – [Legal Information Institute](#)
 - “The Privileges of the Writ of Habeas Corpus shall not be suspended unless when in Cases of Rebellion or Invasion the public Safety may require it.” – [U.S. Constitution](#), Article I, Section 9, Clause 2 (the “Suspension Clause”)
- Requirements for federal habeas relief:
 1. Detainee must be in custody when filed
 2. Detainee must have exhausted all state remedies (where relevant) (see 28 U.S.C. §2241-2256)

Habeas Corpus, continued

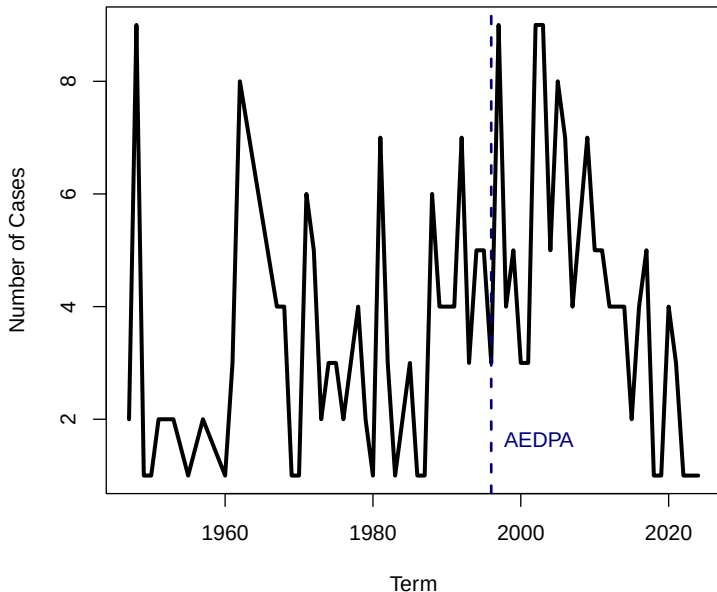
Some history:

- Federal habeas over state detainees established in 1867 (28 USC §2254)
- Suspended at various times...
- Anti-Terrorism and Effective Death Penalty Act (1996) limited federal habeas

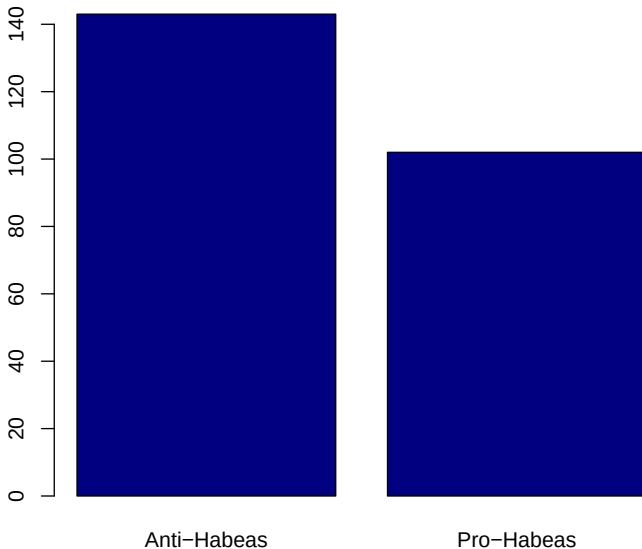
The data:

- Votes in SCOTUS cases addressing habeas corpus (issue = 10020), OT1946-2024 ($N = 246$)
- Coded: 1 = pro-habeas ("liberal"), 0 = anti-habeas ("conservative")

Habeas Corpus Cases Per Term, 1946-2024



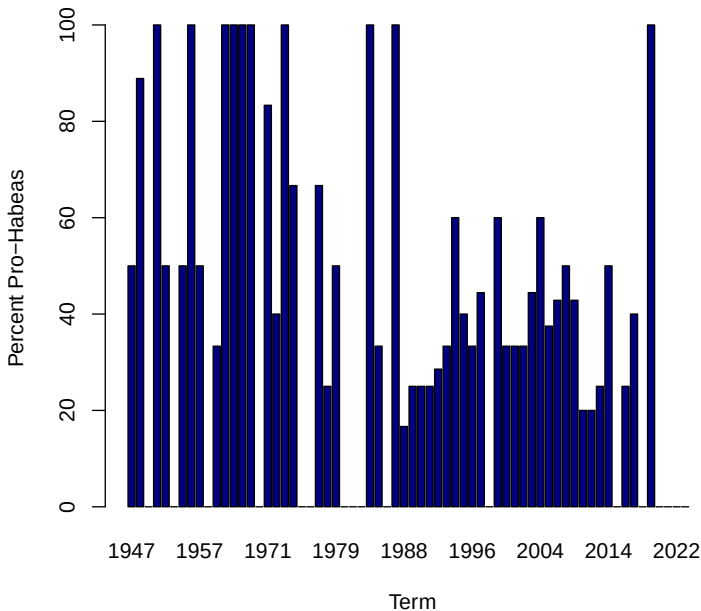
Habeas Corpus Case Outcomes, 1946-2024



Habeas Corpus: Influences

- Judge's Liberalism (Segal-Cover) (expectation: positive)
- Prisoner petition (expectation: negative)
- Does the case involve the application of a federal habeas corpus statute? (expectation: positive)
- Lower Court Disagreement (expectation: positive)
- Post-AEDPA (1996) (expectation: negative)

Habeas Corpus Case Outcomes By Term



Number of Habeas Cases by lawSupp

```
> table(HCases$lawSupp)
```

```
121 207 230 231 314 341 377 400 509 600 900  
  1   1   6   1   2 133   1   5   1  22   4
```

Law	Frequency
Suspension of habeas	1
Fifth Amendment	1
Due Process	6
Equal Protection	1
§1983	2
Federal habeas statutes	133
UMCJ	1
FRAP	5
Treaty law	1
"Infrequently litigated statutes"	22
No legal provision	4

```
# Pro-habeas vote:
```

```
Habeas$ProVote<-Habeas$direction-1
```

```
# Prisoner or defendant is the petitioner:
```

```
Habeas$CrimPet<-ifelse(Habeas$petitioner==126,1,0)           # Prisoner  
Habeas$CrimPet<-ifelse(Habeas$petitioner==215,1,Habeas$CrimPet) # Prisoner  
Habeas$CrimPet<-ifelse(Habeas$petitioner==100,1,Habeas$CrimPet) # Defendant
```

```
# Federal habeas law applicability:
```

```
Habeas$HabLaw<-ifelse(Habeas$lawSupp==341,1,0)
```

```
# Lower court disagreement:
```

```
Habeas$Disagree<-Habeas$lcDisagreement
```

```
# AEDPA:
```

```
Habeas$AEDPA<-ifelse(Habeas$term>1996,1,0)
```

Summary Statistics

```
> Vars<-with(Habeas, data.frame(ProVote,Ideology,CrimPet,  
+                               HabLaw,Disagree,AEDPA))
```

```
> describe(Vars)
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
ProVote	1	2155	0.44	0.50	0.00	0.43	0.00	0	1	1	0.22	-1.95	0.01
Ideology	2	2193	0.42	0.31	0.36	0.40	0.37	0	1	1	0.42	-1.03	0.01
CrimPet	3	2195	0.46	0.50	0.00	0.45	0.00	0	1	1	0.16	-1.98	0.01
HabLaw	4	1584	0.75	0.43	1.00	0.81	0.00	0	1	1	-1.16	-0.65	0.01
Disagree	5	2195	0.28	0.45	0.00	0.22	0.00	0	1	1	0.99	-1.03	0.01
AEDPA	6	2195	0.46	0.50	0.00	0.45	0.00	0	1	1	0.15	-1.98	0.01

Correlations

```
> options(digits=2)
```

```
> cor(Vars,use="complete.obs")
```

	ProVote	Ideology	CrimPet	HabLaw	Disagree	AEDPA
ProVote	1.000	0.3456	0.2770	-0.011	-0.039	-0.090
Ideology	0.346	1.0000	-0.0087	-0.041	-0.032	-0.300
CrimPet	0.277	-0.0087	1.0000	0.107	-0.168	-0.049
HabLaw	-0.011	-0.0411	0.1069	1.000	-0.018	0.051
Disagree	-0.039	-0.0325	-0.1678	-0.018	1.000	0.152
AEDPA	-0.090	-0.3004	-0.0487	0.051	0.152	1.000

Regression Model

$$\begin{aligned} \text{Pro-Habeas Vote}_{ij} = & \beta_0 + \beta_1(\text{Justice Liberalism})_j + \\ & \beta_2(\text{Detainee Petitioner})_i + \\ & \beta_3(\text{Habeas Law Applicability})_i + \\ & \beta_4(\text{Lower Court Disagreement})_i + \\ & \beta_5(\text{AEDPA})_i + u_{it} \end{aligned}$$

Regression Results

```
> Regression <- with(Habeas,lm(ProVote~Ideology+CrimPet+HabLaw+Disagree+AEDPA))
> summary(Regression)
```

Call:

```
lm(formula = ProVote ~ Ideology + CrimPet + HabLaw + Disagree +
    AEDPA)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.949	-0.365	-0.111	0.413	0.931

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.0714	0.0349	2.05	0.041 *
Ideology	0.5905	0.0395	14.96	<2e-16 ***
CrimPet	0.2871	0.0232	12.39	<2e-16 ***
HabLaw	-0.0327	0.0261	-1.25	0.211
Disagree	0.0163	0.0245	0.66	0.507
AEDPA	0.0299	0.0242	1.23	0.218

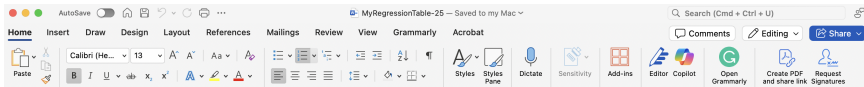
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.44 on 1552 degrees of freedom
(1275 observations deleted due to missingness)

Multiple R-squared: 0.2, Adjusted R-squared: 0.197

F-statistic: 77.5 on 5 and 1552 DF, p-value: <2e-16

Regression Table (MS-Word Version)



1.1. My Regression Table

	Estimate	Standard Error	t value	Pr(> t)	
(Intercept)	0.071	0.035	2.047	0.0408	*
Ideology	0.591	0.039	14.959	0.0000	***
CrimPet	0.287	0.023	12.391	0.0000	***
HabLaw	-0.033	0.026	-1.250	0.2114	
Disagree	0.016	0.024	0.664	0.5066	
AEDPA	0.030	0.024	1.234	0.2175	

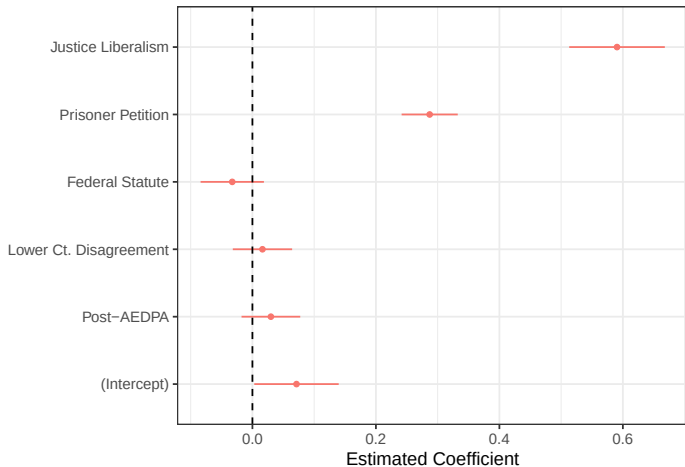
*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

Residual standard error: 0.4436 on 1552 degrees of freedom

Multiple R-squared: 0.1997, Adjusted R-squared: 0.1972

F-statistic: 77.47 on 1552 and 5 DF, p-value: 0.0000

Regression Results: Dotplot



Habeas Corpus Findings

- More liberal justices are substantially more likely to vote in a pro-habeas direction than are conservatives.
 - For example: an 0.5-unit increase in a justice's Segal-Cover score (roughly the equivalent of going from Neil Gorsuch to Ruth Bader Ginsburg) corresponds to an expected $(0.50 \times 0.59 =) 0.295$ increase in the probability of a pro-habeas vote.
- Justices are also more likely to vote in a pro-habeas direction in cases where the detainee is bringing the petition. Such cases have a 0.29 higher probability of a pro-habeas vote, as compared to those in which the state brings the petition.
- We find no conditional association between pro-habeas votes and cases involving a question of the applicability of federal habeas law, where there was lower court disagreement, or in cases decided after the passage of the AEDPA in 1996.